MATH 152 **QUIZ 8** NAME _____

Mrs. Bonny Tighe 25 points

11.8 TA_____Wed 11/9/2022

1. Find the radius of convergence and interval of convergence of the power series.

a) $\sum_{n=1}^{\infty} \frac{(-1)^n \, 5^n \, (x-2)^n}{1 \cdot 3 \cdot 5 \cdots (2n-1)}$

b) $\sum_{k=0}^{\infty} \frac{(-1)^k (x+2)^{2k}}{9^k}$

2. Expand $\sum_{n=0}^{\infty} \frac{(-1)^n (x+1)^{2n+1}}{3^{n+1}(n+3)}$ as a polynomial to the fifth degree.

3. Find the radius of convergence and interval of convergence of the power series.

a) $\sum_{n=1}^{\infty} \frac{(x-2)^n}{3^n\sqrt{n+5}}$

b) $\sum_{n=1}^{\infty} \frac{(2\cdot 5\cdot 8\cdots(3n-1))(x-3)^n}{7^n}$

4. If $\sum_{n=0}^{\infty} c_n x^n$ is convergent when $x=-4$ and divergent when $x=7$, decide which of the following are convergent or divergent . a) $\sum_{n=0}^{\infty} c_n(-3)^n$ _____

b) $\sum_{n=0}^{\infty} c_n$ _____ c) $\sum_{n=0}^{\infty} c_n 9^n$ _____

MATH 152 **QUIZ 8A** NAME _____

Mrs. Bonny Tighe 25 points

11.8 TA_____Wed 11/9/2022

1. Find the radius of convergence and interval of convergence of the power series.

a) $\sum_{n=1}^{\infty} \frac{(1 \cdot 3 \cdot 5 \cdots (2n-1))(3x-1)^n}{5^{n+1}}$

b) $\sum_{k=0}^{\infty} \frac{(x-3)^{2k}}{6^k}$

2. Expand $\sum_{n=0}^{\infty} \frac{(-1)^n (x-1)^n}{4^{n+1}(n+2)}$ as a polynomial to the third degree.

3. Find the radius of convergence and interval of convergence of the power series.

a) $\sum_{n=1}^{\infty} (-1)^n \frac{(x+1)^n}{\sqrt{n+3}}$

b) $\sum_{n=1}^{\infty} \frac{(-1)^n \, 5^n \, (x+2)^n}{1 \cdot 4 \cdot 7 \cdots (3n-2)}$

4. If $\sum_{n=0}^{\infty} c_n x^n$ is convergent when $x = -5$ and divergent when $x = 8$, decide which of the following are convergent or divergent . a) $\sum_{n=0}^{\infty} c_n (-3)^n$ _____

b) $\sum_{n=0}^{\infty} c_n$ _____ c) $\sum_{n=0}^{\infty} c_n 9^n$ _____